A Dependency-Aware Framework for High-Density Storage Infrastructures Under Dynamically Varying Operating Conditions

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Modern high-density data center infrastructures contain dense compute and storage systems where airflow redistribution, pressure fluctuation, structural vibration, and storage performance may interact under dynamically varying operating conditions. This work presents preliminary observations toward a dependency-aware analysis framework for traditional 2U compute nodes and 4U JBOD storage systems. Rolling coupling analysis is employed to characterize evolving interaction behaviors under different operating conditions. Preliminary observations suggest that different infrastructure architectures may exhibit distinct dominant interaction patterns during operational processes.

***Index Terms*—High-density storage infrastructure, operational interaction behavior, rolling coupling analysis, conditional dependency, location-dependent behavior.**

I. Introduction

Modern high-density data center infrastructures contain dense compute and storage systems, where airflow redistribution, pressure fluctuation, structural vibration, and storage performance may interact under dynamically varying conditions. Conventional infrastructure analysis often assumes stationary environments, while local responses may exhibit architecture-dependent dependency behaviors that are difficult to characterize using static approximations [1].

As storage density increases, airflow interaction, vibration propagation, and workload variation become increasingly coupled, potentially leading to location-dependent performance variation across different infrastructure architectures.

This work presents preliminary observations toward a dependency-aware analysis framework for high-density storage infrastructures. The study focuses on traditional 2U compute nodes and 4U JBOD storage systems. Rolling coupling analysis is adopted to evaluate interaction evolution among operational variables and provide insight into architecture-dependent dependency behaviors [1]-[2].

II. DEPENDENCY-AWARE INFRASTRUCTURE FRAMEWORK

This study investigates two representative high-density storage infrastructures: traditional 2U compute nodes and 4U JBOD storage systems. Multiple variables, including workload activity, airflow redistribution, pressure fluctuation, structural vibration, and storage performance are monitored to evaluate evolving interaction behaviors.

Preliminary observations indicate different dominant interaction tendencies across infrastructure architectures. Traditional 2U compute nodes exhibit stronger workload–fan–airflow interactions, whereas 4U JBOD storage systems exhibit more persistent pressure–vibration interactions under dense HDD operating conditions [3].

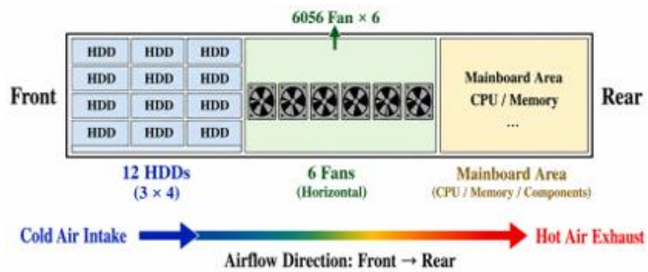

Fig. 1. Internal layout and airflow pathway of a traditional 2U compute node used for dependency-aware operational interaction analysis.

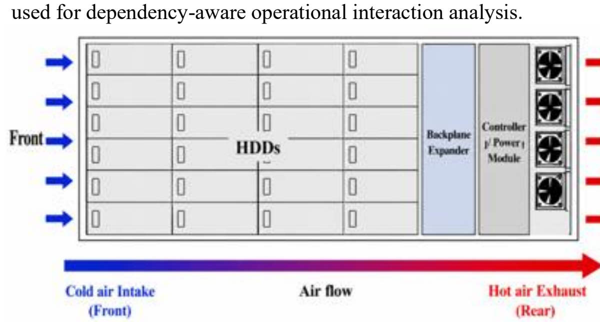

Fig. 2. Internal layout and airflow pathway of a typical 4U JBOD storage infrastructure under dense HDD operating conditions.

III. ROLLING COUPLING ANALYSIS

Rolling coupling analysis is adopted to evaluate evolving dependency behaviors among operational variables. Multiple infrastructure responses, including workload fluctuation, fan speed variation, airflow redistribution, pressure fluctuation, and vibration response are monitored throughout the operational process [1]-[2].

The moving-window correlation coefficient is defined as:

$$r_{xy}^{(k)} = \frac{\sum_{t=k}^{k+W-1}(x_t-\bar{x}^{(k)})(y_t-y^{(k)})}{\sqrt{\sum_{t=k}^{k+W-1}(x_t-\bar{x}^{(k)})^2}\sqrt{\sum_{t=k}^{k+W-1}(y_t-y^{(k)})^2}} \tag{1}$$

Where W denotes the moving window size, and $r_{xy}^{(k)}$ represents the rolling dependency strength between two operational variables within the $(k)-th$ operational interval.

A higher $r_{xy}^{(k)}$ value indicates stronger interaction tendency between two operational responses during a specific operational phase, whereas lower correlation values imply weaker dependency behaviors under dynamically varying operating conditions [2].

A rolling correlation framework is adopted to evaluate evolving dependency behaviors among operational variables using a moving temporal window. Fig. 3 shows dominant workload–fan RPM–airflow interactions in the traditional 2U compute tray system, whereas Fig. 4 shows more persistent pressure–vibration–latency interactions in the typical 4U JBOD storage system.

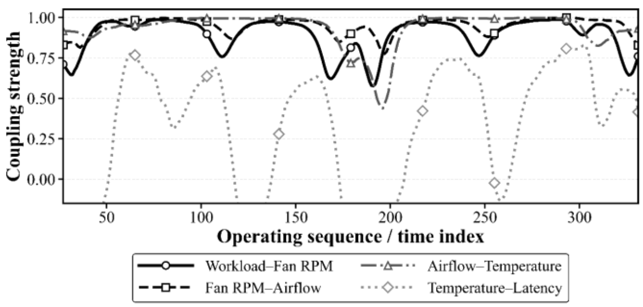

Fig. 3. Rolling dependency evolution observed in the traditional 2U compute tray system under dynamically varying operating conditions.

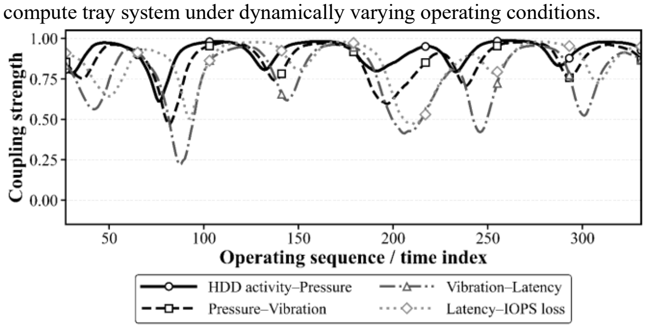

Fig. 4. Rolling dependency evolution observed in the typical 4U JBOD storage system under dense HDD operating conditions.

IV. LOCATION-DEPENDENT OPERATIONAL OBSERVATIONS

A. Preliminary Slot-Level Observations

Preliminary observations are conducted on the traditional 2U compute tray system to investigate location-dependent slot response behaviors under different interaction conditions. Compared with the 4U JBOD storage system, the relatively simplified airflow-driven interaction environment of the 2U architecture provides a more interpretable basis for preliminary dependency analysis and subsequent conditional dependency modeling.

Fig. 5 illustrates the location-dependent slot response distribution under different operational interaction conditions defined in Table I. Elevated response regions become more noticeable under mixed-phase and recirculation-dominant conditions.

Fig. 6 further demonstrates the evolution of normalized slot response indices. Elevated response tendencies become more noticeable in slots 0, 3, 6, and 9 during higher interaction phases, suggesting spatial dependency across the physical slot layout.

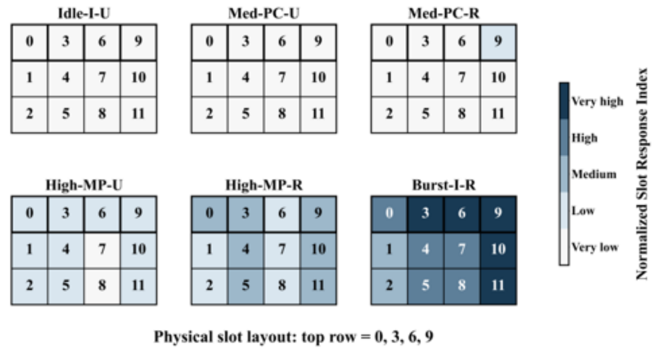

Fig. 5. Location-dependent slot response distribution under different operating conditions in the traditional 2U compute tray system.

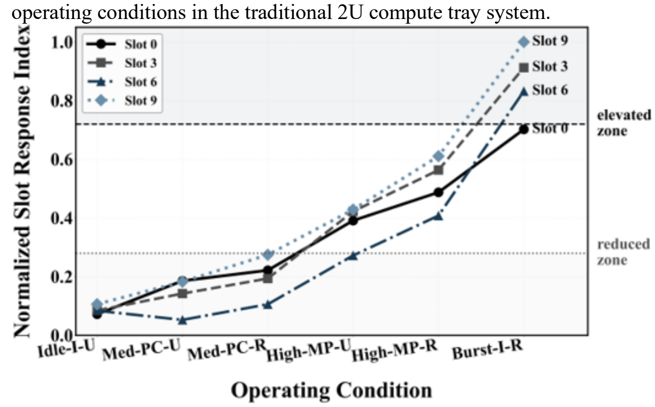

Fig. 6. Evolution of normalized slot response indices under different operating conditions in the traditional 2U compute tray system.

TABLE I
OPERATIONAL INTERACTION CONDITIONS

Symbol	Meaning
I	In-phase fan interaction
PC	Partial-cancel interaction
MP	Mixed-phase interaction
U	Uniform airflow state
R	Recirculation airflow state

B. Concluding Remarks and Future Work

The rolling coupling observations suggest that infrastructure behaviors may not be governed by isolated variables alone. While rolling correlation captures pairwise interaction tendencies, more complex conditional dependency structures may exist across operational responses.

Future work will focus on Graphical Gaussian Model (GGM)-based conditional dependency modeling to investigate infrastructure-level dependency networks under dynamically varying operating conditions [4].

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